

Akshat Navlani

Roll No.: PES2UG23AM009

B.Tech CSE (AI & ML) | PES University, Bengaluru

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EDUCATION

•Bachelor of Technology — Computer Science and Engineering (AI & ML)

Sept 2023 – Present

PES University, Bengaluru

CGPA: 7.83

- Distinction Scholarship — 4 consecutive semesters (3rd through 6th sem); awarded for sustained academic merit
- Relevant Coursework: Data Structures & Algorithms, OOP, Operating Systems, Distributed Systems, Computer Networks, DBMS, Software Engineering, AI & Machine Learning

TECHNICAL SKILLS

Languages: Java, C, C++, Python, Go (Golang), JavaScript, TypeScript, SQL, Zig, Verilog

Frameworks & Libraries: React, Next.js, React Native, Expo, Express.js, Gin, Node.js, FastAPI, Astro, Tailwind CSS, shadcn/ui, GSAP

AI / ML: PyTorch, TensorFlow, Keras, HuggingFace Transformers, LangChain, LangGraph, Ollama, FAISS, OpenCV, Gemini API, Groq API, Reinforcement Learning

Databases: PostgreSQL, MongoDB, MySQL, SQLite, Redis, Firebase Firestore

DevOps & Cloud: Docker, GitHub Actions, CI/CD Pipelines, Nginx, Vercel, Railway, Render, Git, Linux, Bash

Testing & Tools: Playwright (E2E), Jest, Vitest, pytest, Postman, Figma

CS Fundamentals: Data Structures & Algorithms, System Design, OOP, REST APIs, Concurrency, DBMS, Computer Networks, OS

PROJECTS

•Multimodal PDF Summarizer with XAI Attribution

Python | HuggingFace Transformers | PyTorch | FAISS | BLIP-2 | Gemini API | Groq API | Gradio | pytest

Repo

- Designed an end-to-end multimodal AI pipeline that extracts and processes text, tables, figures, and charts from scientific PDFs, generating sentence-level XAI explainability scores with per-modality contribution ratios — enabling transparent, auditable summarisation across all content types.
- Integrated five interchangeable LLM backends (Gemini, Groq, BART, extractive, hybrid) behind a unified interface with FAISS-backed semantic retrieval; evaluated throughput and output quality against ROUGE, BERTScore, and BLEU benchmarks, exposing backend switching without code changes.
- Deployed the pipeline as a Gradio web application with full pytest test coverage, reducing upload-to-summary latency through asynchronous chunk indexing and lazy model loading.

•Personalized News Emailer — Production-Grade Full-Stack Platform

React 18 | Vite | Express.js | Node.js | Firebase Auth | Firestore | Playwright | Jest | Vitest

Repo

- Built a full-stack personalised news platform with Firebase Auth, 12-category RSS aggregation, a 15-minute in-memory caching layer to reduce redundant upstream fetches, and automated Handlebars-templated email delivery with per-user preference routing.
- Enforced 75%+ test coverage via GitHub Actions CI across 415 Playwright E2E tests and Jest/Vitest unit suites; eliminated unnecessary re-renders by applying `memo`, `useMemo`, and `useCallback` across critical render paths, measurably reducing component update frequency.
- Implemented WCAG 2.1 AA accessibility compliance, session-timeout handling, dark-mode theming via CSS variables, and an Express monitoring middleware layer — shipping a scalable, observable, and accessible production web application.

•Airbnb Price Prediction — End-to-End ML Pipeline

Python | XGBoost | LightGBM | scikit-learn | pandas | NumPy | Jupyter

Repo

- Developed a complete ML pipeline on 26,000 Airbnb listings — data cleaning, feature engineering (10+ custom features: amenity binary flags, review frequency, availability ratios), log-normalised targets, and early stopping — training an XGBoost regression model achieving $R^2=0.82$ and $MAE=\$66.59$.
- Constructed a companion 3-class price-tier classifier with quantile-based labelling and a full scikit-learn preprocessing pipeline (RobustScaler + OneHotEncoder), reaching 97.54% accuracy and $F1=0.975$; packaged the final model as a reusable pipeline artifact ready for REST API serving via a single `predict()` call.

•SAE ADDC Autonomous Drone — Vision-Guided Navigation System

Python | DroneKit | MAVLink | OpenCV | ArUco | pyzbar | NumPy

Repo

- Developed a competition-grade autonomous UAV for SAE ADDC integrating real-time QR-code target tracking, ArUco-marker precision landing, and MAVLink/DroneKit flight control with a servo-actuated payload delivery mechanism.
- Eliminated targeting error caused by lens distortion by performing full intrinsic camera calibration (cv2.calibrateCamera) on a 20-image checkerboard dataset, improving drop-zone detection reliability at competition altitude and enabling sub-pixel-accurate visual servoing.

•Film Production Database — Relational Schema & RBAC Engine

MySQL | SQL DDL/DML | Stored Procedures | Triggers | Window Functions | draw.io

Repo

- Implemented a 20+ entity normalised relational database in MySQL, centralising data integrity across 14 audit triggers and 12 CHECK constraints and reducing manual validation overhead by shifting business-rule enforcement to the database layer.
- Provisioned a 3-tier role-based access control (RBAC) system granting real MySQL user privileges (admin/manager/viewer via GRANT); encoded profit, ROI, payroll, and box-office analytics as stored functions with 15+ stored procedures and 8 views, documented via ER and relational schema diagrams.

ACHIEVEMENTS & CERTIFICATIONS

•1st Place — IEEE Codefusion Hackathon

PES University

- Designed and delivered a secure distributed application with optimised graph-traversal algorithms for high-speed data processing; selected as best overall solution by an industry judging panel over competing university teams.

•3rd Place — CoDMAV CodeJam Hackathon

- Built a scalable AI-integrated application under time-constrained conditions, combining ML inference with a robust, horizontally scalable software architecture.

•Distinction Scholarship — PES University

3rd – 6th Semester

- Awarded for sustained academic excellence; CGPA 7.83 at end of third year (6th semester).

•Certifications

- Postman API Fundamentals Expert — REST API development, testing, collection automation, and integration workflows.
- HackerRank Certified: C Programming | Data Structures & Algorithms.

LEADERSHIP & POSITIONS OF RESPONSIBILITY

•Club Head — CodeChef PES EC Campus Chapter

2023 – Present

- Scaled a competitive programming community from scratch to 100+ active members; organised and executed 3 college-wide programming contests with combined attendance exceeding 1,000 participants, managing end-to-end event logistics, problem-setting, and judging coordination.
- Ran weekly DSA training sessions and mock technical interview drives covering arrays, graphs, dynamic programming, and system design — directly preparing members for software engineering recruiting pipelines at product companies.

•Club Head — PAWS (Campus Animal Welfare Society)

2023 – Present

- Directed welfare operations for 20+ stray dogs across the PES University campus, coordinating feeding schedules, veterinary visits, and shelter improvements in collaboration with university administration.

•Member — Avions Aviation & Drone Club

2023 – Present

- Contributed embedded systems and computer vision expertise to competition-grade autonomous UAV builds; served as a technical contributor on the drone navigation system at SAE ADDC.